18.02 Practice Exam 3 A

1. Let $(\bar{x}, \bar{y})$ be the center of mass of the triangle with vertices at $(-2, 0)$, $(0, 1)$, $(2, 0)$ and uniform density $\delta = 1$.

a) (10) Write an integral formula for $\bar{y}$. Do not evaluate the integral(s), but write explicitly the integrand and limits of integration.

b) (5) Find $\bar{x}$.

2. (15) Find the polar moment of inertia of the unit disk with density equal to the distance from the y-axis.

3. Let $\vec{F} = (ax^2y + y^3 + 1)\mathbf{i} + (2x^3 + bxy^2 + 2)\mathbf{j}$ be a vector field, where a and b are constants.

a) (5) Find the values of a and b for which $\vec{F}$ is conservative.

b) (5) For these values of a and b, find $f(x, y)$ such that $\vec{F} = \nabla f$.

c) (5) Still using the values of a and b from part (a), compute $\int_C \vec{F} \cdot d\vec{r}$ along the curve C such that $x = e^t \cos t$, $y = e^t \sin t$, $0 \le t \le \pi$.

4. (10) For $\vec{F} = yx^3\mathbf{i} + y^2\mathbf{j}$, find $\int_C \vec{F} \cdot d\vec{r}$ on the portion of the curve $y = x^2$ from $(0, 0)$ to $(1, 1)$.

5. Consider the region R in the first quadrant bounded by the curves $y = x^2$, $y = x^2/5$, $xy = 2$, and $xy = 4$.

a) (10) Compute $dxdy$ in terms of $dudv$ if $u = x^2/y$ and $v = xy$.

b) (10) Find a double integral for the area of R in uv coordinates and evaluate it.

6. a) (5) Let C be a simple closed curve going counterclockwise around a region R. Let $M = M(x, y)$. Express $\oint_C M dx$ as a double integral over R.

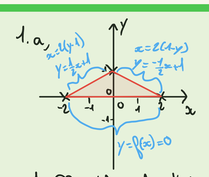
b) (5) Find M so that $\oint_C M dx$ is the mass of R with density $\delta(x, y) = (x + y)^2$.

7. Consider the region R enclosed by the x-axis, $x = 1$ and $y = x^3$.

a) (5) Use the normal form of Green's theorem to find the flux of $\vec{F} = (1 + y^2)\mathbf{j}$ out of R.

b) (5) Find the flux out of R through the two sides C_1 (the horizontal segment) and C_2 (the vertical segment).

c) (5) Use parts (a) and (b) to find the flux out of the third side C_3.



1. a) $\bar{y} = \frac{1}{\text{Aire}} \iint_R y\,dA$ Aire d'un triangle = base × hauteur × $\frac{1}{2}$

Aire = $4 \times 1 \times \frac{1}{2} = 2$

$\bar{y} = \frac{1}{2} \int_0^1 \int_{2y-2}^{2(1-y)} y\,dx\,dy$

b) L'axe y est un axe de symétrie donc $\bar{x}$ doit s'y trouver, donc $\bar{x} = 0$.

4. $\vec{F} = yx^3\mathbf{i} + y^2\mathbf{j}$ $\quad y = x^2$ de $(0,0)$ à $(1,1)$

Intégrale large avec $x = t$ donc $y = t^2$; $dx = dt$; $dy = 2t$

$\vec{F} \cdot d\vec{r} = M\,dx + N\,dy$ $\quad M = yx^3$

$N = y^2$

$\vec{F} \cdot d\vec{r} = yx^3\,dx + y^2\,dy$ $\quad M\,dx = t^2 t^3\,dt = t^5\,dt$

$N\,dy = t^4 \cdot 2t = 2t^5$

$\int \vec{F} \cdot d\vec{r} = \int_0^1 (t^5 + 2t^5)\,dt = \int_0^1 (3t^5)\,dt$

$= \int_0^1 3\frac{t^5}{6} = \left[3\frac{t^6}{6}\right]_0^1 = \left[\frac{t^6}{2}\right]_0^1 = \left[\frac{1}{2} - 0\right] = \frac{1}{2}$

2. Moment d'inertie polaire : mesure la résistance à la rotation autour de l'origine

$I_0 = \iint_R x^2 \cdot \delta\,dA$

δ = distance à l'axe y

Coordonnées polaires : $x = r\cos\theta \Rightarrow |x| = r|\cos\theta| = \delta$

r va de 0 à 1

θ va de 0 à 2π

$I_0 = \iint_R r^2 \cdot r|\cos\theta| \cdot r\,dr\,d\theta$

$|\cos\theta|$ est positif sur $[0, \pi/2]$ et négatif sur $[\pi/2, \pi]$, etc. Mais par symétrie, l'intégrale sur $[0, 2\pi]$ avec $|\cos\theta|$ vaut exactement **4 fois** l'intégrale sur $[0, \pi/2]$ où $\cos\theta \ge 0$ (donc pas besoin de valeur absolue).

$I_0 = 4\int_0^{\pi/2}\int_0^1 r^4 \cos\theta\,dr\,d\theta$

$= 4\int_0^{\pi/2}\int_0^1 \frac{s}{5}\cos\theta\,ds\,d\theta$

$= 4\int_0^{\pi/2}\left[\frac{s}{5}\cos\theta\right]_0^1 d\theta = 4\int_0^{\pi/2}\left[\frac{1}{5}\cos\theta - 0\right]d\theta$

$= 4\int_0^{\pi/2} \frac{1}{5}\sin\theta = 4\left[\frac{1}{5}\sin\left(\frac{\pi}{2}\right) - \frac{1}{5}\sin(0)\right] = 4\left[\frac{1}{5} \cdot 1 - \frac{1}{5} \cdot 0\right] = \frac{4}{5} = I_0$

5. $y = x^2$, $y = \frac{x^2}{5}$, $xy = 2$, $xy = 4$

a) $u = \frac{x^2}{y}$ $\quad v = xy$

$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{2x}{y} & -\frac{x^2}{y^2} \\ y & x \end{vmatrix}$ $\quad \det = \left[\frac{2x}{y}\right]x - \left[-\frac{x^2}{y^2}\right]y$

$= \frac{2x^2}{y} + \frac{x^2}{y} = \frac{3x^2}{y}$

$du\,dv = \frac{3x^2}{y}\,dx\,dy$

$\Leftrightarrow \frac{y}{3x^2}\,du\,dv = dx\,dy$ $\quad u = \frac{x^2}{y} \Leftrightarrow \frac{1}{u} = \frac{y}{x^2}$

$\Leftrightarrow \frac{1}{3u}\,du\,dv = dx\,dy$

b) $\begin{vmatrix} u = \frac{x^2}{y} \Leftrightarrow y = x^2 \\ v = \frac{x^2}{5} = 1 \end{vmatrix}$ $\begin{vmatrix} u = \frac{x^2}{y} \quad y = \frac{x^2}{5} \\ u = \frac{x^2}{x^2/5} \Leftrightarrow \frac{x^2}{5} = 5 \end{vmatrix}$ $\begin{vmatrix} v = xy = 2 \end{vmatrix}$ $\begin{vmatrix} v = xy = 4 \end{vmatrix}$

$\begin{cases} 1 \le u \le 5 \\ 2 \le v \le 4 \end{cases}$

$\iint_R dx\,dy \Leftrightarrow \text{Aire} = \int_2^4\int_1^5 \frac{1}{3u}\,du\,dv = \int_2^4\int_1^5 \frac{1}{3}\ln(u)\,dv = \int_2^4 \left[\frac{1}{3}\ln(u)\right]_1^5 dv = \int_2^4 \left[\frac{1}{3}\ln(5) - \underbrace{\frac{1}{3}\ln(1)}_{=0}\right]dv = \int_2^4 \left[\frac{1}{3}\ln(5) - 0\right]dv = \int_2^4 \frac{1}{3}\ln 5\,dv = \left[\frac{1}{3}\ln 5\,v\right]_2^4 = \left[\frac{4}{3}\ln 5 - \frac{2}{3}\ln 5\right] = \frac{2}{3}\ln 5$

6. Green's Theorem : $\oint_C M dx + N dy = \iint_R (N_x - M_y)\,dA$

Ici : $N dy = 0$ donc : $\oint_C M dx = \iint_R -M_y\,dA$

$\delta = (x+y)^2$

Masse = $\iint_R (x+y)^2\,dA$

On cherche M_y tq $-M_y = (x+y)^2$

$(x+y)^2 = x^2 + 2xy + y^2$

$M = x^2y + xy^2 + \frac{y^3}{3} \Leftrightarrow -M = -x^2y - xy^2 + \frac{y^3}{3}$

7. $\vec{F} = \overbrace{(1+y^2)}^{N}\mathbf{j}$ $\quad M = 0$

Flux = $\iint_R \text{div}\,\vec{F}\,dA$

$\text{div}\,\vec{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}$

$\text{div}\,\vec{F} = \frac{\partial N}{\partial y} \Leftrightarrow \text{div}\,\vec{F} = \frac{\partial(1+y^2)}{\partial y} \Leftrightarrow \text{div}\,\vec{F} = 2y$

Flux = $\iint_R 2y\,dA$

Région R, délimitée par :

- $y = 0$
- $x = 1$
- $y = x^3$

$\iint_R 2y\,dA = \int_0^1\int_0^{x^3} 2y\,dy\,dx$

$= \int_0^1\int_0^{x^3} \frac{2y^2}{2}\,dx = \int_0^1 \left[y^2\right]_0^{x^3} dx = \int_0^1 \left[x^6 - 0\right]dx = \int_0^1 \frac{x^7}{7} = \left[\frac{x^7}{7}\right]_0^1 = \frac{1}{7}$

Flux à travers une courbe = $\int_C \vec{F} \cdot \hat{n}\,ds$

Sur C_1 ; $(y = 0)$; $\hat{n} = -\hat{\jmath}$

$\vec{F} \cdot \hat{n} = (1+y^2)\hat{\jmath} \cdot (-\hat{\jmath}) = -\hat{\jmath}\,(1+y^2)$

Sur C_1 $y = 0$ donc $\vec{F} \cdot \hat{n} = (-1)$

On intègre une constante sur un intervalle, donc ça équivaut à : constante × longueur de l'intervalle

Soit : $(-1) \cdot 1 = -1$ $\quad \text{Flux}_{C_1} = (-1)$

C_2 : $(1,0)$ à $(1,1)$

$\vec{F} \cdot \hat{n} = (1+y^2)\hat{\jmath} \cdot \hat{\imath} = 0$ $\quad \hat{\imath}$ et $\hat{\jmath}$ sont perpendiculaire ici, donc $\hat{\jmath} \cdot \hat{\imath} = 0$

Flux total = $\frac{1}{7}$

$\underbrace{C_1}_{-1} + \underbrace{C_2}_{0} + C_3 = \frac{1}{7}$ donc $C_3 = \frac{1}{7} + 1 \Leftrightarrow C_3 = \frac{8}{7}$

3. a) $\vec{F} = (ax^2y + y^3 + 1)\mathbf{i} + (2x^3 + bxy^2 + 2)\mathbf{j}$

$M = ax^2y + y^3 + 1$ $\quad M_y = ax^2 + 3y^2$

$N = 2x^3 + bxy^2 + 2$ $\quad N_x = 6x^2 + by^2$

Pour $\vec{F}$ conservatif on a besoin de $M_y = N_x \Leftrightarrow ax^2 + 3y^2 = 6x^2 + by^2$

$\Leftrightarrow a = 6$, $b = 3$

b) On cherche $\vec{F}$ tq $\vec{F} = \nabla f$, donc $f_x = M$, $f_y = N$

$f_x = \int ax^2y + y^3 + 1\,dx = a\frac{x^3}{3}y + xy^3 + x + c(y) = 6\frac{x^3}{3}y + xy^3 + x + c(y) = 2x^3y + xy^3 + x + c(y)$

On sait que $N = 2x^3 + bxy^2 + 2$ donc on doit avoir $2x^3 + bxy^2 + 2 = \frac{\partial}{\partial y}\left(2x^3y + xy^3 + x + c(y)\right)$

$\Rightarrow 2x^3 + bxy^2 + 2 = 2x^3 + 3xy^2 + c(y)$

$\Leftrightarrow 2x^3 + 3xy^2 + 2 = 2x^3 + 3xy^2 + c(y)$

$= 2 \Leftrightarrow c(y) = 2y$ (+ constante)

$f(x,y) = 2x^3y + xy^3 + x + 2y$

c) $\int_C \vec{F} \cdot d\vec{r}$ avec $x = e^t\cos t$ et $y = e^t \sin t$ avec $0 \le t \le \pi$

$t = 0$: $x = e^0\cos 0 = 1 \cdot 1$ $\quad$ Départ : $(1,0)$

$y = e^0 \sin 0 = 1 \cdot 0$ $\quad$ Arrivée : $(-e^\pi, 0)$

$t = \pi$: $x = e^\pi \cos\pi = e^\pi \cdot (-1)$

$y = e^\pi \sin\pi = e^\pi \cdot 0$

Puisque $\vec{F}$ est conservatif on a $\int_C \vec{F} \cdot d\vec{r} = f(\text{Arr. vée}) - f(\text{Départ})$

$= f(-e^\pi, 0) - f(1,0)$

$= \left[2(-e^\pi)^3 0 + (-e^\pi)^3 0^2 + (-e^\pi) + 2 \cdot 0\right] - \left[2 \cdot 1^3 0 + 1 \cdot 0^3 + 1 + 2 \cdot 0\right]$

$= \left[0 + 0 - e^\pi + 0\right] - \left[0 + 0 + 1 + 0\right]$

$= -e^\pi - 1$